Meanders and Stuff

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Nikolai J. Friedberg

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Kyle Ormsby

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I want to thank a few people.

Preface

This is an example of a thesis setup to use the reed thesis document class.

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Abstract

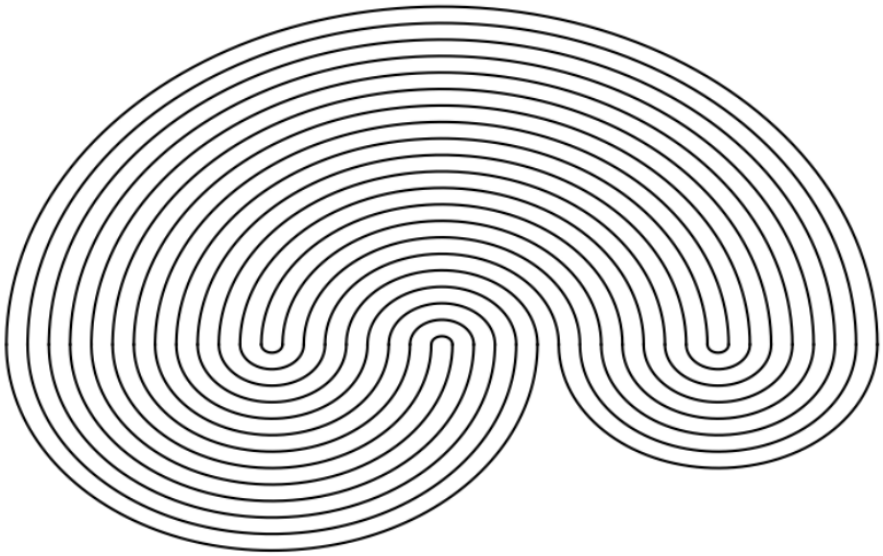
The preface pretty much says it all.

Dedication

You can have a dedication here if you wish.

Introduction

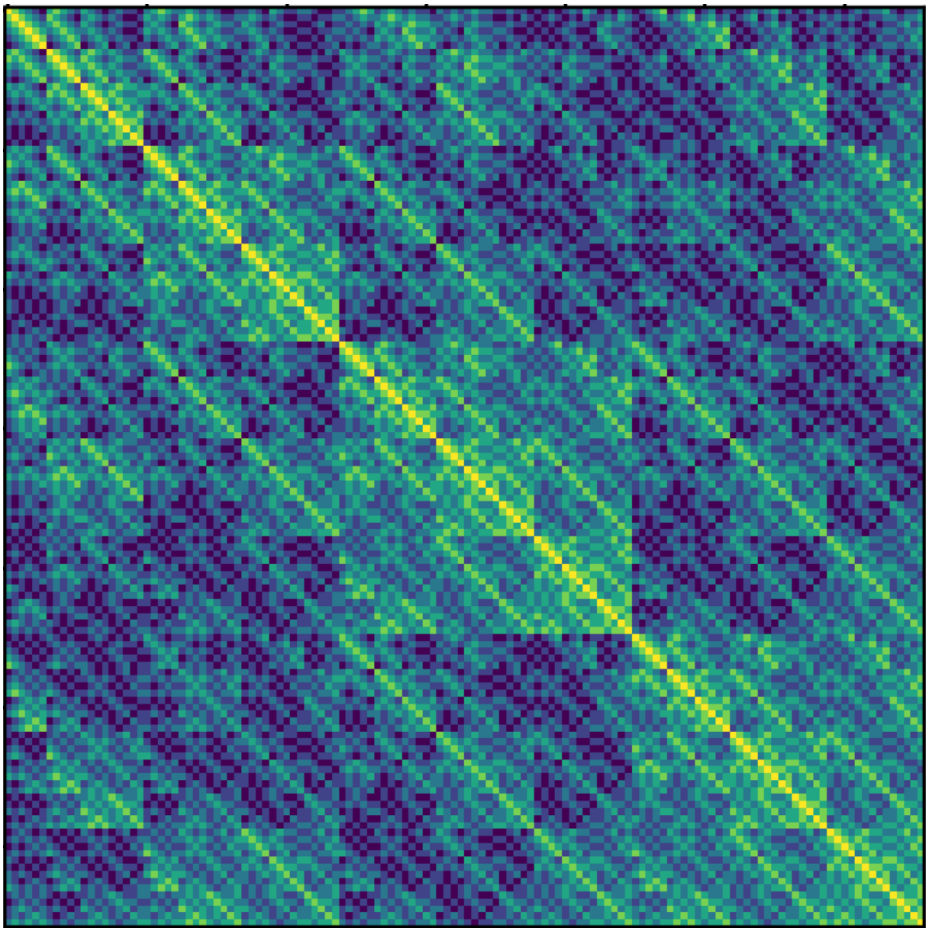
Coming all the way from the Greek *Maiandros*, referring to a winding river in what is now Turkey, we derive our modern English word "meander". This has come to take on several meanings, such as the winding course of a river or a persons aimless walk. More recently, it has also come to denote a certain mathematical object consisting of a line and a closed curve meandering over it, crossing it at certain intersection points. Despite this simplicity, or perhaps because of it, these objects can be strikingly beautiful.



Also despite their simplicity, they remain mathematically elusive. The question of how many meanders exist of a certain size has resisted mathematicians for decades and remains open. Further, as they have been studied more, meanders and their patterns have been shown to pertain to important problems in theoretical physics (see for example Zvonkin 2024). This is perhaps unsurprising if one takes the view of meanders as 2D knots, or as we will see, 2D links (knots with multiple components).

In this paper, we approach the study of meanders from an experimental mathematics perspective, finding and studying patterns that emerge in meandric systems. We also re-exposit an important proof in the meander literature. In chapter 1, we introduce and explore several key objects involved in the study of meanders. Chapter 2 contains a reexposition of a paper by Di Francesco on the Meander matrix and its associated determinant, which reveal some of the deeper implications of Meander patterns. We focus on ironing out certain details that were obfuscated. Chapter 3

contains our most novel contributions, wherein we analyze the aforementioned Meander matrix. We find an intriguing link between meanders and Narayana numbers and study the emergence of interleavings of Catalan structures within meandric systems. We also provide a codebase useful for studying meandric systems and their patterns.



Chapter 1

Meanders

1.1 Meanders and their constituents (and their constituents' equivalences)

A meander consists of a single self-avoiding closed curve that intersects a line finitely many times. Meanders can also be considered to be non-bifurcating labyrinths (Zvonkin 2024). As the number of points of intersection must be even (if it goes down, it must come back up), we denote the order of a meander to be $2n$, the number of points of intersection. The graph of the curve itself does not matter in our classification: only the number of points of intersection. The meander can be separated into two configurations, one above and one below the line, each of which consists of n "arches" connecting the $2n$ points as pairs. Only very particular combinations of arch configurations form single closed curves. Generalizing, we look at meandric systems of order $2n$ which can contain multiple individual meanders.

A major question in the study of meandric systems is deciding how many meandric systems are composed of just a single meander. This remains as of yet unanswered.

Now that we have introduced meandric systems and their constituent arch diagrams, we will now also introduce a series of important equivalences with these two

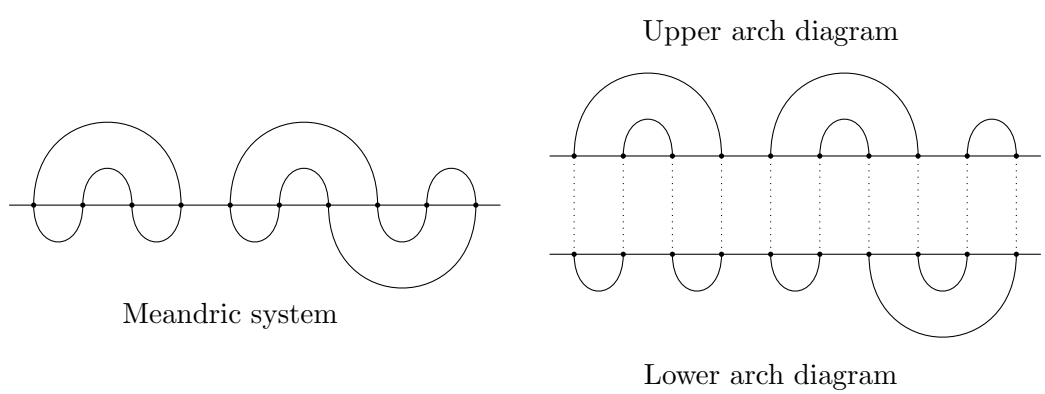


Figure 1.1: An order 5 meandric system with 2 connected components (left), and its constituent arch diagrams (right)

which we will proceed to shift between somewhat freely depending on which construct best suits our needs.

1.1.1 Arch Diagrams

Generally, we consider these arches to be in the set

$$A_{2n} = \{\text{set of arch configurations of length 2n}\}.$$

Arch configurations can be viewed as a series of non-crossing connections between $2n$ points on a line, where all the connections are above the line. Naturally, one wonders the number $|A_{2n}|$ of arch configurations of length $2n$.

First, note that since every point is connected and the arches are noncrossing, there must be an even number of points under any given arch. Now, consider a connection from point 1 to point $2j$ for $1 \leq j \leq n$. This divides the line into sections $[2, 2j-1]$ and $[2j+1, 2n]$, which themselves can be viewed as $A_{2(j-1)}$ and $A_{2(n-j)}$. Thus, with $|A_0| = 1$, we have

$$|A_{2n}| = \sum_{1 \leq j \leq n} |A_{2(j-1)}||A_{2(n-j)}| = \frac{(2n)!}{(n+1)!n!} = c_n$$

where c_n is the nth Catalan number (see Pak 2014 for a lovely treatment of Catalan numbers and their history).

1.1.2 Arch Diagram Permutations and their Use

Arch diagrams can also naturally be represented by permutations. For an arch diagram μ, the permutation σ_μ is built of a series of two-cycles denoting the position of the start and end of each arch. For example, the upper arch diagram in Figure 1.1 would be denoted (1 4)(2 3)(5 8)(6 7)(9 10).

Now, let μ, ℓ be arch diagrams and let $\#\pi_o\left(\frac{\mu}{\ell}\right)$ denote the number of closed curves, or separate connected components, of the meander formed by connected upper arch diagram μ with lower arch diagram ℓ.

Proposition 1.1. $\#\pi_o\left(\frac{\mu}{\ell}\right) = \frac{\#\text{ of cycles in } \sigma_\ell^{-1}\sigma_\mu}{2}$

First, let us look at the example of Figure 1.1. Denoting the top arch diagram as μ and the bottom as ℓ, we have

$$\sigma_\mu = (1\ 4)(2\ 3)(5\ 8)(6\ 7)(9\ 10), \quad \sigma_\ell = (1\ 2)(3\ 4)(5\ 6)(7\ 10)(8\ 9)$$
$$\sigma_\ell^{-1}\sigma_\mu = (1\ 3)(2\ 4)(5\ 9\ 7)(6\ 10\ 8)$$

As we can see, there are 4 cycles in $\sigma_\ell^{-1}\sigma_\mu$ and 2 connected components in the meandric system $\frac{\mu}{\ell}$.

More generally, we can first note that every component of the meandric system is always of even length since a closed curve cannot be closed on the same side as you start. Then, since we traverse $\frac{\mu}{\ell}$ in steps of 2 through the lens of the permutation multiplication, every component of the meandric system is cut in half, resulting in twice the number of cycles as there are connected components. □

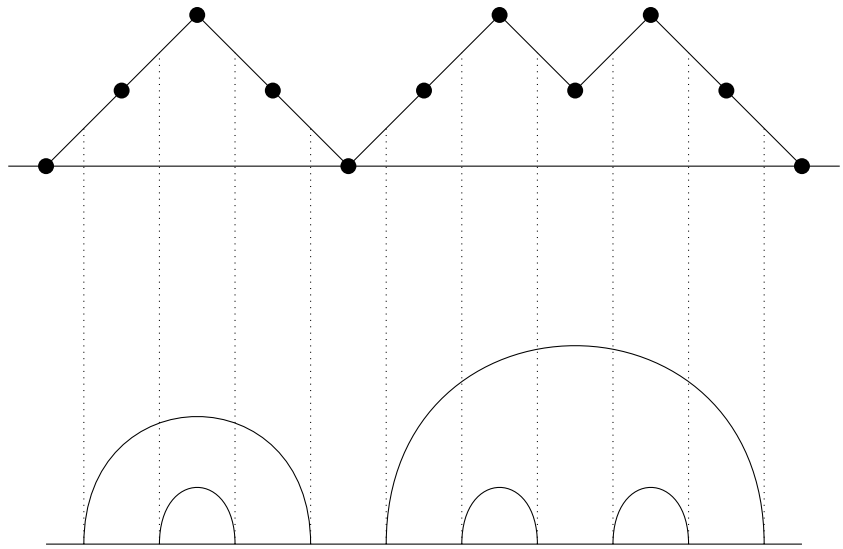


Figure 1.2: A closed walk and its corresponding arch diagram

1.1.3 Walks

Closed Walks

For our first construction equivalent to the arch configurations, let us consider W_{2n}, closed walk diagrams of length $2n$, also known as Dyck words. These are formed by $2n$ steps either going up and to the right or down and to the left, and never dipping below it's starting height. Denoting these steps u and d respectively, for a walk diagram to be closed, we must have that $\#u = \#d$.

A bijection between A_{2n} and W_{2n} can be seen by taking any element $A \in A_{2n}$ and constructing a walk w with a u positioned wherever an arch begins and a d wherever an arch ends. One can also see that there is a unique solution to the inverse series of operations when constrained to noncrossing arches, as you replace an u step with the beginning of an arch and end the arch at the next d step of the same height as the beginning of the arch. This establishes the bijection, and so we also have that $|A_{2n}| = |W_{2n}|$.

Open Walks

As a complementary structure to open arch diagrams, we introduce $W_n^{(h)}$, the set of open walks of final height h. Similarly, u steps are placed where arches begin and d steps are placed where they end. As such, every time an open arch begins, the final height increases by one, and so the final height equals the number of open arches. We define the height i to be the number of arches above the point, plus the number of open arches to the left of i. With this bijection we have

$$|W_n^{(h)}| = |A_n^{(h)}| = c_{n,h} \tag{1.2}$$

1.1.4 Open arch configurations

For purposes that will soon be clear, we also introduce

$$A_n^{(h)} = \{\text{set of arch configurations of length n with h open arches}\}.$$

Open arch configurations have some number of arches which never connect back down to another point, ostensibly shooting off to "infinity". To count these objects, we consider the arch coming from the first point, and consider the following two cases:

1. If the arch is open, the rest of the arch configuration is some general arch configuration $a \in A_{n-1}^{(h-1)}$.

2. If the arch connects to some bridge $2j$, the arch configuration is then separated into an ordinary arch configuration of order $2(j-1)$ under the arch, and an open arch configuration $a \in A_{n-2j}^{h}$ (order $n-2j$ with h open arches).

Thus, we have

$$|A_n^h| = |A_{n-1}^{h-1}| + \sum_{j=1}^{\lfloor n/2 \rfloor} |A_n||A_{n-2j}^h|.$$

With the initial condition of $|A_{2n}^{(0)}| = c_n$, this recursion can be summarized by the expression

$$|A_n^{(h)}| = c_{n,h} = \binom{n}{\frac{n-h}{2}} - \binom{n}{\frac{n-h}{2}-1} \tag{1.3}$$

Note that $A_n^{(h)}$ is only defined if $n = h \mod 2$.

1.2 Temperley-Lieb Algebra

Another equivalence can be made with reduced elements of the Temperley-Lieb Algebra $TL_n(q)$, and here is one area where meandric systems have some "practical" application. In the following exposition, certain words will be placed in quotation marks to indicate that there is a large amount of knowledge that we do not claim to fully understand, and is further elucidated in associated citations. The Temperley-Lieb algebra first arose in the 1970s in connection with problems in statistical mechanics regarding hydrogen bonding lattices, specifically, in the context of the "transfer matrix method applied to the Potts model and ice-type models" (Doty and Giaquinto 2025). By studying the vector space of Temperley-Lieb elements, we are, by extension, studying the "representation" of the algebra. If the determinant of the Gram matrix of the Temperley-Lieb algebra, which we will be studying later in this paper, is non-zero, then the algebra is "semisimple" (Citation=Zajj?). Through this connection, this seemingly abstract combinatorial objects are shown to reflect something about the nature of our world.

The algebra itself can best be thought of as $\mathbb{C}$-linear combinations of "string diagrams". These diagrams consist of a series of n strings connecting $2n$ points, with

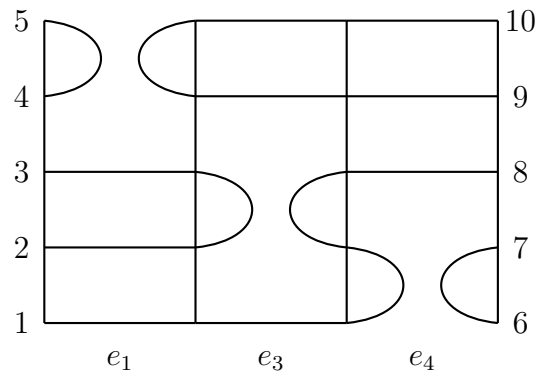


Figure 1.3: A element $e \in TL_5$ resulting from composition of generators $e_1e_3e_4$.

the first n points listed on a left line from top to bottom and the latter n on a right line listed from bottom to top. Each point is connected to exactly one other point, and the strings cannot cross. A product ef between two elements is taken by putting the two diagrams next to each other, connecting the right ends of the strings of e with the left ends of the strings of f.

Formally, $TL_n(q)$ is defined by the n generators $1, e_1, e_2, \ldots, e_{n-1}$ and a set of relations. The identity element 1 is portrayed by a diagram connecting each point on the left to its corresponding point on the right with a straight string, and e_i is the same except with point i connected to point $i+1$ and point $n-i$ connected to point $n-i+1$ (diagramatically, these are placed directly opposite each other). Combined with these generators, the following relations completely describe the algebra:

(i) $e_i^2 = qe_i$

(ii) $e_ie_j = e_je_i$ if $|i-j| > 1$

(iii) $e_ie_{i\pm1}e_i = e_i$

Relation (i) says that if a closed circle is made in the product of elements, it can be removed and accounted for by a factor of q. Relation (ii) simply says that if two strings are far apart, they commute. Relation (iii) says that different looking strings that start and end in the same place are the same, and so unnecessary turns in the string can be removed by "pulling" a string straight.

An element $e \in TL_n$ is said to be reduced if all loops have been removed (relation (i)), all strings have been pulled straight (relation (iii)), and it has been written as a minimal product of the generator elements e_i (using all three relations).

The equivalence between A_{2n} and reduced elements of TL_n can be seen visually as a sort of "opening up" of the Temperley-Lieb element, connecting the bottoms of the two lines of points and keeping the connecting strings as they are, simply reinterpreting them now as arches.

A further equivalence can be made to an ideal $\mathcal{I}_n(q)$ of $TL_{2n}(q)$ generated by the element $u_n = e_1e_3 \ldots e_{2n-1}$. Pictorially, we make this equivalence simpler by turning diagrams of elements on their side, so that the top is made of n strings connecting

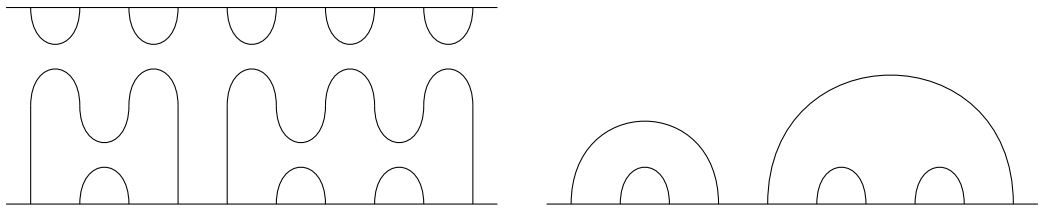


Figure 1.4: An element of $\mathcal{I}_{2n}$ and it's corresponding arch diagram

the upper n pairs of points, and the strings connecting the points on the bottom are a "wiggly" arc diagram of order $2n$.

Chapter 2

Meander Determinant

2.1 Meanders and their Equivalent Constituents

In the following chapter, we will draw extensively from the existing elucidation of the proof of the formula for the meander determinant published by Di Francesco. Our purpose is to re-expound the proof in a more approachable manner.

2.1.1 Meander Determinant

We now introduce the main theorem of this chapter. Recalling the previous definitions from Chapter 1, we introduce the meander matrix $\mathcal{G}_{2n}(q)$, where q is any complex number. [1]

Definition 2.1 ($\kappa(a|b)$)**.** $\kappa(a|b)$ will denote the number of connected components of the meandric system resulting from arch diagrams a and b (see Figure 1.1).

With this, the entries of the meander matrix are given by

$$\left[\mathcal{G}_{2n}(q)\right]_{a,b} = q^{\kappa(a|b)} \qquad a, b \in A_{2n}. \tag{2.2}$$

Now, we introduce polynomials which we will call Di Francesco polynomials, denoted by $U_m(q)$. They are defined by

$$U_m(2\cos\theta) = \frac{\sin(m+1)\theta}{\sin\theta} \tag{2.3}$$

for $m \geq 0$. Note that these are essentially Chebyshev polynomials of the second kind, though since Di Francesco uses a nonstandard normalization, we elect to denote them differently so as not to confuse the reader. Di Francesco polynomials can be written in terms of the standard Chebyshev polynomials of the second kind as $U_m^{DF}(q) = U_m^{standard}(q/2)$. For intuition as to how this trigonometrically defined function is a polynomial at all, recall that trigonometric identities can be used to rewrite the

[1] q is often equivalently construed as a formal variable, but for our purposes we use a complex number, though the particular one chosen has no particular meaning.

right side of the equation purely in terms of cos, resulting in a polynomial in $\cos\theta$. Replacing $\cos\theta$ with q, this can be summarized in the recurrence relation

$$U_0(q) = 1, \qquad U_1(x) = q, \qquad U_{n+1}(q) = xU_n(q) - U_{n-1}(q)$$

With this definition, as well as the integer valued numbers $c_{n,m}$ defined above, we assert the following formula for the determinant of the meander matrix $\mathcal{G}_{2n}(q)$:

Theorem 2.4 (Closed formula for determinant of the meander matrix)**.**

$$\det\left[\mathcal{G}_{2n}(q)\right] = \prod_{m=1}^{n} \left[U_m(q)\right]^{a_{2n,2m}}$$

$$a_{2n,2m} = c_{2n,2m} - c_{2n,2m+2}$$

This result will at this point appear completely unmotivated, and will be elucidated in the pages to come. However, that there is a closed formula for the determinant, yet not for the entries of the determinant itself, hints at something profound about patterns in meandric systems.

2.1.2 A New Tool: Boxes

Now we will introduce a map ρ from W_{2n}, the set of walks of length $2n$, to reduced elements of the ideal $\mathcal{I}_n(q)$ of $TL_{2n}(q)$. To begin with, let us consider the fundamental walk, denoted a_n. This consists of an up step and a down step repeated $n/2$ times, such that for any $0 \leq i \leq n/2 - 1$,

$$h(2i) = 0 \text{ and } h(2i+1) = 1$$

where h is a height function. The map ρ takes this walk to the reduced element

$$\rho(a_n) = u_n = e_1 e_3 ... e_{2n-1}$$

Definition 2.5 (Box Additions)**.** A box addition on a minimum i of any walk diagram $a \in W_{2n}$ consists taking $h(i) \to h(i) + 2$, filling the minimum and taking it to a maximum. Diagramatically, this looks like adding a square box rotated 45°. We let $(a + \beta_i)$ denote a box addition on $a \in W_{2n}$.

From this, we can build any reduced element by defining a box addition on the fundamental walk by

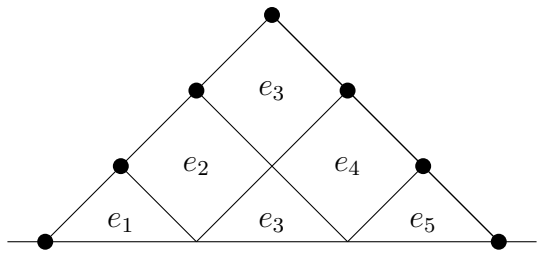
$$\rho(a + \beta_i) = e_i \rho(a).$$

Diagrammatically, a box addition β_i takes a minimum at i in a to a maximum, taking $h(i) \to h(i) + 2$, increasing the height by 2.

Therefore, we write a general element $e \in \mathcal{I}_n(q)$ as a product over all e_i's corresponding to the necessary box additions to build from a_n to a such that $\rho(a) = e$. For example, in TL_5, the walk that looks like a pyramid can be written as

$$((a_n + \beta_2) + \beta_4) + \beta_3 = (e_2 a_n + \beta_4) + \beta_3) = e_4 e_3 a_n + \beta_3 = e_3 e_4 e_2 a_n = e_3 e_4 e_2 e_1 e_3 e_5$$

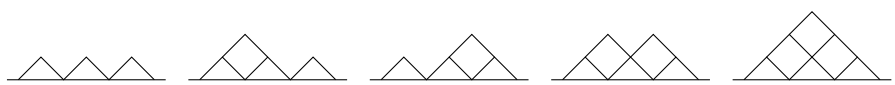
Further, we can define the length of a walk diagram $a \in W_{2n}$ to be the number of box additions needed to construct a from a_n.



2.1.3 Vector Space of $\mathcal{I}_n(q)$

From these reduced elements of $\mathcal{I}_N(q)$ we can construct a basis for the corresponding vector space over the complex numbers. Further, we know that $|W_{2n}| = c_n = \dim(\mathcal{I}_n(q))$. We will refer to this basis as the reduced basis, denoted B_R. For concision, when referring to to the basis elements as walks, we will denote them $(a)_1$. As an example, the reduced basis for $\mathcal{I}_3(q)$ is given by the following $c_3 = 5$ elements

$$\{e_1e_3e_5, e_2e_1e_3e_5, e_4e_1e_3e_5, e_2e_4e_1e_3e_5, e_3e_2e_4e_1e_3e_5\} \tag{2.6}$$



Diagrams showing these elements

To translate back to the Temperley-Lieb algebra from these box addition walks, simply let the corresponding arch diagram be the bottom half of an element $e \in \mathcal{I}_n(q)$, and let the top half be the strings connecting the n pairs of upper points as described above.

2.2 Meander Determinant

First, we define the transpose e^t of any $e \in TL_n$ to be the result of swapping the labels of points i and $2n - i$, essentially rotating the diagram 180°.

The Temperley-Lieb Algebra is equipped with a trace operation

Diagram showing what the trace is doing

Definition 2.7 (Trace of Temperley-Lieb Algebra)**.** $\mathrm{Tr}(e) = q^{\kappa(e)}$

where $\kappa(e)$ is the number of connected components after adding strings connecting opposite points i and $n - i + 1$.

Now, noting that $e_i^t = e_i$ and $(ef)^t = f^te^t$, we introduce the bilinear form

$$(e, f) = \mathrm{Tr}(ef^t). \tag{2.8}$$

Using this, we can consider the Gram matrix of the reduced basis of $I_n(q)$ formed by running over the walk diagrams $a, b \in W_{2n}$, where

$$[\Gamma_{2n}(q)]_{a,b} = ((a)_1, (b)_1) = \mathrm{Tr}((a)_1(b)_1^t) \tag{2.9}$$

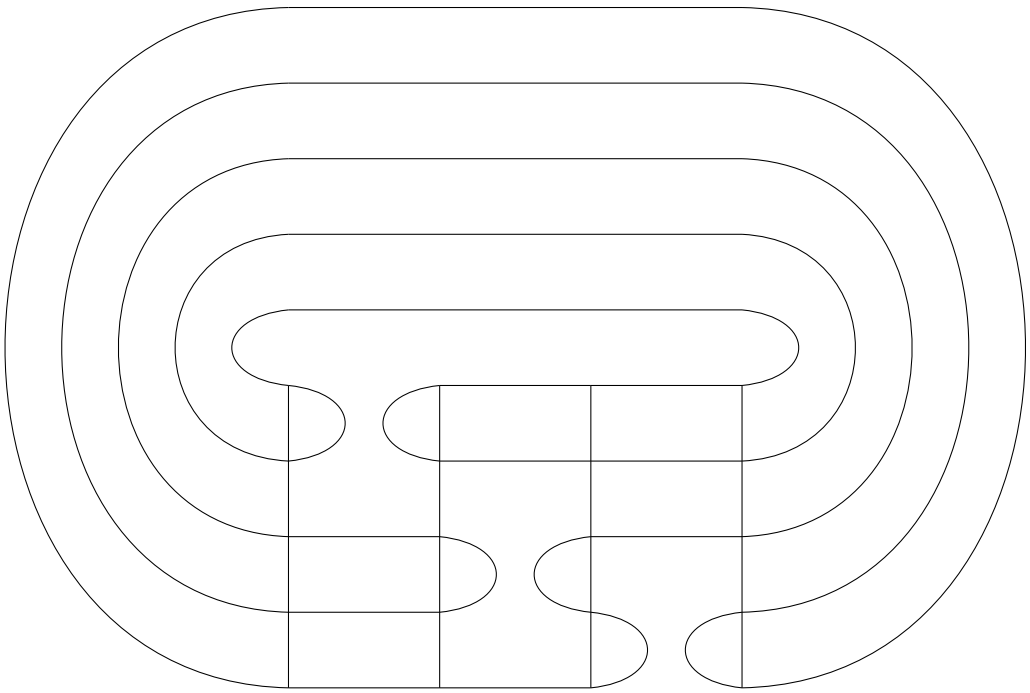


Figure 2.1: A Temperley Lieb diagram with trace 2

The entries are then given by $q^{n+\kappa(a|b)}$, where the n is present because of the linearity of trace, and comes from the n connections of the factors of $u_n = e_1 e_3 \dots e_{2n-1}$ in the $I_n(q)$ representations of both a and b, forming n closed circles (see relation 1 of the Temperley Lieb algebra). The $\kappa(a|b)$ is the standard result given by the trace, connecting ends of strings on the top and bottom of the $I_n q$ diagram. In this case, $q^{\kappa(a|b)}$ represents the number of connected components of the meandric system corresponding to $(a|b) = (a)_1 (b)_1^t = ef^t$.

Now, recalling the definition of the meander matrix $\mathcal{G}_{2n}(q)$ where each entry is the number of connected components of the meandric system, we have that

$$[\Gamma_{2n}(q)]_{a,b} = q^{n+\kappa(a|b)} = q^n [\mathcal{G}_{2n}(q)]_{a,b}.$$

With this, we also have that

$$\det(\Gamma_{2n}(q)) = q^{ncn} \det \mathcal{G}_{2n}(q). \tag{2.10}$$

Thus, calculating the Gram determinant of the reduced basis B_R will also yield the meander determinant.

2.3 Creating a new orthonormal basis

To help us compute the Gram determinant of the reduced basis, we will perform an explicit Gram-Schmidt orthonormalization of the reduced basis in terms of the above bilinear form. This orthonormalization is a change of basis from B_R to a new basis,

which we will call the orthonormal basis, or B_O. We will denote basis elements as walks by $(a)_2$. The orthonormal basis, as one might expect, satisfies the following properties

1. Orthonormal basis elements are still indexed in the same way as reduced basis elements, denoted as $(a)_2$ for $a \in W_{2n}$.

2. Orthonormal basis elements are orthogonal, meaning that $((a)_2, (b)_2) = 0$ when $a \neq b$.

3. Orthonormal basis elements all have norm 1, meaning that $((a)_2, (a)_2) = 1$ for $a \in W_{2n}$.

To begin the construction of our orthonormal basis elements, we start with the fundamental element $(a_n)_2$, similar to the reduced basis but with a new normalization factor introduced.

$$(a_n)_2 = q^{-n} e_1 e_3 ... e_{2n-1} \tag{2.11}$$

This normalization factor is necessary for property 3, such that

$$((a_n)_2, (a_n)_2) = \mathrm{Tr}((a_n)_2 (a_n)_2^t) = \mathrm{Tr}((a_n)_2) = q^{-n} q^n = 1$$

as n internal loops create a q^n to cancel out one of the two q^{-n}s, and n external loops from the trace cancel out the other q^{-n}.

Now, to continue the constrution of the orthonormal basis elements, we return to box additions, this time with a new definition. A box addition takes place at index i and height ℓ (recall that the base walk consists of heights $0, 1, 0, 1, \ldots, 0, 1, 0$, and that a box addition occurs at a minimum, increasing the height by 2). Then, we have that

$$(a + \beta_{i,\ell})_2 = \sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}} (e_i - \mu_\ell)(a)_2 \tag{2.12}$$

where we denote μ_ℓ as

$$\mu_\ell = \frac{U_{\ell-1}(q)}{U_\ell(q)} \quad \text{for } \ell \geq 1 \tag{2.13}$$

where U_m is a Chebyshev polynomial of the second kind. Together with $(a_n)_2$, the above box addition rule completely defines the orthonormal basis elements. The recursion relation

$$U_{m+1}(q) = qU_m(q) - U_{m-1}(q)$$

now turns into

$$\frac{1}{\mu_1} - \mu_m = \frac{1}{\mu_{m+1}} \tag{2.14}$$

This new box addition for the orthonormal basis can be viewed as analogous to the box additions for the reduced basis, but with some new baggage in order to ensure the basis elements remain orthonormal. For instance, the $\sqrt{\mu_{\ell+1}/\mu_\ell}$ factor ensures the basis elements remain normal, though this is certainly not obvious. However, the algebra simply works out, as will soon be seen.

Note since all of these elements are built on the factor $e_1 e_3 \dots e_{2n-1}$, they all belong to the ideal $\mathcal{I}_n(q)$. Therefore, B_O elements can be expressed in terms of B_R elements with μ_ℓ coefficients as follows:

$$(a)_2 = \sum_{b \subset a} P_{b,a(b)_1} \tag{2.15}$$

Here, each entry $P_{b,a}$ is the μ-coefficients of the front of the expanded product created by all the orthonormal basis box additions needed to create a from b. More specifically, $P_{b,a}$ are the coefficients in front of the B_R representation of b. For example, the change of basis matrix for $c_3 = 5$ looks like

Diagrams showing the which walks correspond to which row/column

$$P_3 = \begin{bmatrix} \mu_1^3 & -\mu_1^{7/2}\mu_2^{1/2} & -\mu_1^{7/2}\mu_2^{1/2} & \mu_1^4\mu_2^1 & -\mu_1^4\mu_2^{3/2}\mu_3^{1/2} \\ 0 & \mu_1^{5/2}\mu_2^{1/2} & 0 & -\mu_1^3\mu_2^1 & \mu_1^3\mu_2^{3/2}\mu_3^{1/2} \\ 0 & 0 & \mu_1^{5/2}\mu_2^{1/2} & -\mu_1^3\mu_2^1 & \mu_1^3\mu_2^{3/2}\mu_3^{1/2} \\ 0 & 0 & 0 & \mu_1^2\mu_2 & -\mu_1^2\mu_2^{3/2}\mu_3^{1/2} \\ 0 & 0 & 0 & 0 & \mu_1^2\mu_2^{1/2}\mu_3^{1/2} \end{bmatrix} \tag{2.16}$$

In Dyck path and box diagram notation, the orthonormal basis elements of $\mathcal{I}_3(q)$ look like

$$\begin{aligned} &\mu_1^3 e_1 e_3 e_5 \\ &\mu_1^{5/2}\mu_2^{1/2}(e_2 - \mu_1) e_1 e_3 e_5 \\ &\mu_1^{5/2}\mu_2^{1/2}(e_4 - \mu_1) e_4 e_1 e_3 e_5 \\ &\mu_1^2\mu_2(e_2 - \mu_1)(e_4 - \mu_1) e_2 e_4 e_1 e_3 e_5 \\ &\mu_1^2\mu_2^{1/2}\mu_3^{1/2}(e_3 - \mu_2)(e_2 - \mu_1)(e_4 - \mu_1) e_3 e_2 e_4 e_1 e_3 e_5 \end{aligned}$$

The expression resulting from expanding one of these expressions corresponds to a column of P. For example,

$$\mu_1^{5/2}\mu_2^{1/2}(e_2 - \mu_1) e_1 e_3 e_5 = -\mu_1^{7/2}\mu_2^{1/2} e_2 e_1 e_3 e_5 + \mu_1^{5/2}\mu_2^{1/2} e_1 e_3 e_5$$

2.4 Proving Orthogonality and Normality

Proposition 2.17. The orthonormal basis is orthonormal with regards to the bilinear form, namely

$$((a)_2, (b)_2) = \delta_{a,b} \qquad \text{for all } a, b \in W_{2n}.$$

Following Di Francesco, we will first recurse on box additions to prove the following:

2.4.1 Lemma 1.

$$(a)_2^t(b)_2 = 0 \qquad \text{for all } a, b \in W_{2n} \text{ such that } |a| \leq |b| \text{ and } a \neq b$$

In fact, this lemma is a stronger result than we need, but does imply the orthogonality required. In the following computations, we will define the length of a B_O element $(a)_2$, $|a|$, to be the number of B_O box additions needed to make $(a)_2$ from the fundamental walk $(a_n)_2$. We will now prove Lemma 1 by recursion on B_O box addition. First, suppose that the property described in Lemma 1 holds for some $(a)_2$. Now, we will show that it holds for any B_O box addition at some minimum of $(a)_2$, written as $(a)_2 + \beta_i$. Namely, we show that for $b \geq |a + \beta_i|$,

$$(a + \beta_{i,\ell})_2^t(b)_2 = 0 \tag{2.18}$$

The trick here is to transfer the box addition from a to b such that the inductive hypothesis can again apply. Note that since Temperley-Lieb generators are self adjoint, so are B_O box additions. With this in mind, we have

$$\begin{aligned}(a + \beta_{i,\ell})_2^t(b)_2 &= \left(\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i - \mu_\ell)(a)_2\right)^t (b)_2 \\ &= (a)_2^t\left(\left(\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i - \mu_\ell\right)(b)_2\right),\end{aligned}$$

at which point the aforementioned property will again apply.

However, when transferring the B_O box addition, there are three possible outcomes in b:

1. b has a minimum at i with $h(i) = m$.
2. b has a maximum at i with $h(i) = m$.
3. b has a slope at i with $h(i) = m$.

note that this height m is liable to be different from the height ℓ at which the box addition would occur in a. With this in mind, our goal here is to, after transferring the box addition over, express $\beta_{i,\ell}(b)_2$ as a linear combination of box additions on walk diagrams b', all of which have $|b'| \geq |b| - 1$ so that our inductive hypothesis can take effect on each term, reducing the whole thing to 0.

First, we deal with the most natural case, where b also has a minimum at i at height $m - 1$ and the box transfer takes place at height m. We can write the B_O box addition as

$$\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i - \mu_\ell) = \sqrt{\frac{\mu_{\ell+1}\mu_m}{\mu_\ell\mu_{m+1}}} \times \sqrt{\frac{\mu_{m+1}}{\mu_m}}(e_i - \mu_m) + \sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(\mu_m - \mu_\ell)$$

where in the first term we have introduced a box addition at point i and height m. Part of this is canceled out in the leading coefficient, and part of it is canceled in the second term. With this, we can write the box addition on b as

$$\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i - \mu_\ell)(b)_2 = \sqrt{\frac{\mu_{\ell+1}\mu_m}{\mu_\ell\mu_{m+1}}}(b + \beta_{i,m})_2 + \sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(\mu_m - \mu_\ell)(b)_2 \tag{2.19}$$

leaving us with a linear combination as desired.

Let us now deal with case 2, where b has a maximum at i at height m. Here, b is the result of a previous box addition at i on the walk $b - \beta_{i,m}$, where previously there was a minimum at i with $h(i) = m - 1$. Therefore, we can write the basis 2 box addition of $(b)_2$ as

$$\begin{aligned}
\beta_{i,\ell}(b)_2 &= \sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i - \mu_\ell)(b)_2 \\
&= \sqrt{\frac{\mu_{\ell+1}\mu_{m+1}}{\mu_\ell\mu_m}}(e_i - \mu_\ell)(e_i - \mu_m)(b - \beta_{i,m})_2 \\
&= \sqrt{\frac{\mu_{\ell+1}\mu_{m+1}}{\mu_\ell\mu_m}}\left[(e_i^2 - \mu_\ell e_i - \mu_m e_i + \mu_\ell\mu_m)\right](b - \beta_{i,m})_2
\end{aligned}$$

Here, we use the property $e_i^2 = qe_i = \mu_1^{-1}e_i$, and add and subtract terms $\mu_1^{-1}\mu_m$ and μ_m^2, resulting in

$$\begin{aligned}
\beta_{i,\ell}(b)_2 &= \sqrt{\frac{\mu_{\ell+1}\mu_{m+1}}{\mu_\ell\mu_m}}\left[(\mu_1^{-1} - \mu_\ell - \mu_m)(e_i - \mu_m) + \mu_m(\mu_1^{-1} - \mu_m)\right](b - \beta_{i,m})_2 \\
&= \sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(\mu_{m+1}^{-1} - \mu_\ell)(b)_2 + \sqrt{\frac{\mu_{\ell+1}\mu_m}{\mu_\ell\mu_{m+1}}}(b - \beta_{i,m})_2
\end{aligned}$$

where in this last line we have used identity 2.14 twice, and thus have a linear combination as desired.

The final case to reckon with is when b has a slope at i at height m. First, note that we can write the B_O box addition on $(b)_2$ as

$$\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i - \mu_\ell)(b)_2 = \left(\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}e_i - \sqrt{\mu_\ell\mu_{\ell+1}}\right)(b)_2 \tag{2.20}$$

resulting in a first term proportional to a reduced basis box addition and a constant in the second term. Then, we can write out $(b)_2$ as a series of previous B_O box additions. Using this, a B_R box addition on a slope, in this case ascending, looks like

$$\begin{aligned}
e_i(b)_2 &= e_i\sqrt{\frac{\mu_{m+1}}{\mu_m}}\sqrt{\frac{\mu_m}{\mu_{m-1}}}(e_{i+1} - \mu_m)(e_i - \mu_{m-1})\cdots \\
&= e_i\sqrt{\frac{\mu_{m+1}}{\mu_{m-1}}}(e_{i+1}e_i - e_{i+1}\mu_{m-1} - e_i\mu_m + \mu_m\mu_{m+1})\cdots \\
&= \sqrt{\frac{\mu_{m+1}}{\mu_{m-1}}}\left(e_i(1 - \mu_m(\mu_1^{-1} + \mu_{m-1})) - \mu_{m-1}e_ie_{i+1}\right)\cdots \\
&= \sqrt{\frac{\mu_{m+1}}{\mu_{m-1}}}\left(e_i(1 - \mu_m\frac{1}{\mu_m}) - \mu_{m-1}e_ie_{i+1}\right)\cdots \qquad \text{using 2.14} \\
&= -\sqrt{\mu_{m+1}\mu_{m-1}}e_ie_{i+1}\cdots
\end{aligned}$$

Here, we are left with a factor proportional to e_i acting on a reduced basis addition (factor e_{i+1}) acting on the rest of the orthonormal basis box additions of $(b)_2$. Therefore, we can repeat the above calculation with $i \to i+1$ and $m \to m-1$ over and over

until we reach the bottom of the diagram. Here, we are left with a B_R box addition e_{i+m-1} acting on orthonormal basis box addition $\beta_{i+m,1}$, which itself is acting on the fundamental walk made of B_R boxes, which includes factor e_{i+m-1}. Therefore, the calculation is reduced to

$$\begin{aligned} e_i(b)_2 &= \cdots e_{i+m-1}\sqrt{\frac{\mu_2}{\mu_1}}(e_{i+m}-\mu_1)e_{i+m-1}\cdots \\ &= \cdots \sqrt{\frac{\mu_2}{\mu_1}}(e_{i+m-1}e_{i+m}e_{i+m-1}-\mu_1(e_{i+m-1}^2))\cdots \\ &= \cdots \sqrt{\frac{\mu_2}{\mu_1}}(e_{i+m-1}-\mu_1(\mu_1^{-1}e_{i+m-1}))\cdots \\ &= 0 \end{aligned}$$

using Temperley-Lieb relations 1 and 3. Thus, any orthonormal basis box addition on a slope of $(b)_2$ reduces to

$$\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i-\mu_\ell)(b)_2 = -\sqrt{\mu_\ell\mu_{\ell+1}}(b)_2 \tag{2.21}$$

leaving us with a linear combination as desired.

In all three cases 1.20-1.22, we can then rewrite

$$(a+\beta_{i,\ell})_2^t(b)_2 = (a)_2^t\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i-\mu_\ell)(b)_2 = \sum_{b'\in W_{2n}}\lambda_{b'}(a)_2^t(b')_2 \tag{2.22}$$

This linear combination consists of terms $(a)_2^t(b')_2$. Now, note that $b' \in \{b, b+\beta_{i,\ell}, b-\beta_{i,\ell}\}$, so by our initial assumption that $b \geq |a+\beta_i|$, we have $|b'| \geq |a|$. Therefore, we can apply the recursion hypothesis to each of terms of the linear combination, which must then all vanish. With this, we have established $(a+\beta_{i,\ell})_2^t(b)_2 = 0$, as long as the product involving $(a)_2$ itself vanishes. Now to complete the recursion, we now prove the base case for the fundamental walk a_n, namely that

$$(a_n)_2^t(b)_2 = 0 \qquad \text{for all } b \in W_{2n} \text{ such that } b \neq a_n. \tag{2.23}$$

This amounts to evaluating $e_1e_3\ldots e_{2n-1}(b)_2$, namely, n reduced basis box additions on b. However, since $b \neq a$, b must consist of at least one B_O box additions. We have already shown, in case 3, that a B_R box addition on a slope of B_O box additions vanishes. Thus, the above product also vanishes, and so our base case is verified. Lemma 1 follows by recursion.

Now, we prove Proposition 1. For any $a, b \in W_{2n}$ we have that by symmetry of the bilinear form, $((a)_2,(b)_2) = ((b)_2,(a)_2)$. Thus, if $a \neq b$, we can apply Lemma 1 to (a,b) if $|a| \leq |b|$ or to (b,a) if $|a| \geq |b|$, resulting in a zero result in either case. Now, we work through the case $a = b$, and show that orthonormal basis elements have norm 1. We already have that the fundamental walk has norm 1 (2.11) Now, we show that this property persists for any B_O box addition, using the same trick of

transferring the box addition.

$$\begin{aligned}
(a+\beta_{i,\ell})_2^t(a+\beta_{i,\ell})_2 &= (a)_2^t\frac{\mu_{\ell+1}}{\mu_\ell}(e_i-\mu_\ell)^2(a)_2\\
&= (a)_2^t\frac{\mu_{\ell+1}}{\mu_\ell}(e_i^2-2e_i\mu_\ell+\mu_\ell^2)(a)_2\\
&= (a)_2^t\frac{\mu_{\ell+1}}{\mu_\ell}(e_i(\mu_1^{-1}-2\mu_\ell)+\mu_\ell^2)(a)_2\\
&= (a)_2^t\frac{\mu_{\ell+1}}{\mu_\ell}(e_i(\mu_{\ell+1}^{-1}-\mu_\ell)+\mu_\ell^2)(a)_2\\
&= (a)_2^t(e_i\mu_\ell^{-1}-e_i\mu_{\ell+1}+\mu_\ell\mu_{\ell+1})(a)_2
\end{aligned}$$

Here, we remember Lemma 1 2.4.1. If we can get the above algebra in the form of $(a)_2^t(a)_2$ plus something that Lemma 1 applies to, then we have our result. Thus, we use the trick of adding and subtracting the same thing. In this case, we add and subtract 1, getting

$$\begin{aligned}
&= (a)_2^t(1+e_i\mu_\ell^{-1}-1-e_i\mu_{\ell+1}+\mu_{\ell+1}\mu_\ell)(a)_2\\
&= (a)_2^t\left(1+\frac{\mu_{\ell+1}}{\mu_\ell}\left(e_i\mu_{\ell+1}^{-1}-\mu_{\ell+1}^{-1}\mu_\ell-e_i\mu_\ell+\mu_\ell^2\right)\right)(a)_2\\
&= (a)_2^t(a)_2+\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(\mu_\ell-\mu_{\ell+1}^{-1})\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(e_i-\mu_\ell)(a)_2^t(a)_2\\
&= (a)_2^t(a)_2+\sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}(\mu_\ell-\mu_{\ell+1}^{-1})(a)_2^t(a+\beta_{i,\ell})_2\\
&= (a)_2^t(a)_2
\end{aligned}$$

2.5 Proof of Meander Determinant Formula

Now that we have derived shown that our orthonormal basis is in fact orthonormal, let us proceed in calculating the Gram determinant for the reduced basis. Since B_O is orthonormal, its Gram matrix is the identity matrix I of dimension c_n. Therefore, the change of basis from B_R to B_O looks like

$$P\Gamma_{2n}(q)P^t = I \tag{2.24}$$

and so $\det\Gamma_{2n}(q) = (\det P)^{-2}$. Recall that our change of basis matrix P is upper triangular. This yields the formula

$$\det P = \prod_{a\in W_{2n}} P_{a,a}$$

where only the diagonal elements enter the computation. From the definition of a B_O box addition, we see that the diagonal elements satisfy[2]

$$P_{a+\beta_{i,\ell},a+\beta_{i,\ell}} = \sqrt{\frac{\mu_{\ell+1}}{\mu_\ell}}P_{a,a}. \tag{2.25}$$

[2]See 2.16.

With the initial condition for the fundamental walk diagram of $P_{a_n,a_n} = \mu_1^n$, this recursion completely defines the diagonal elements $P_{a,a}$. Therefore, we have

$$P_{a,a}^2 = \mu_1^{2n} \prod_{\text{boxes of } a} \frac{\mu_{\ell+1}}{\mu_\ell}. \tag{2.26}$$

Then, notice that the boxes of a can be arranged into n strips where each strip is increasing in height and position. Let us define the length of a strip to be the height of the uppermost box in the strip. For example, the fundamental walk is composed of n strips of length 1, and by adding the first possible box at $i = 2$, the length of the first strip is increased by one. Using this, we can rewrite the above formula as

$$P_{a,a}^2 = \mu_1^n \prod_{\text{strips of } a} \mu_m \tag{2.27}$$

wherein we have canceled n of the μ_1s with the bottom of all of the strips, and the rest of the μ's cancel out as the product moves up the strip, leaving only the final $\mu_{\ell+1} = \mu_m$ with m being the height of the strip. Then, using our previous formula for the Gram determinant of the redued basis, we have

$$\begin{aligned} \det \Gamma_{2n}(q) &= (\det P)^{-2} \\ &= \prod_{a \in W_{2n}} P_{a,a}^{-2} \\ &= \mu_1^{-nc_n} \prod_{\substack{\text{strips of} \\ \text{all } a \in W_{2n}}} \mu_m^{-1} \end{aligned}$$

With this, and recalling our previous formula 2.10, we have

$$\det \mathcal{G}_{2n}(q) = \prod_{\substack{\text{strips of} \\ \text{all } a \in W_{2n}}} \mu_m^{-1} \tag{2.28}$$

which can further be rewritten as

$$\det \mathcal{G}_{2n}(q) = \prod_{m=1}^{n} [\mu_m]^{-s_{2n,m}} \tag{2.29}$$

where $s_{2n,m}$ denotes the number of strips of length m across all walks in W_{2n}. Now, to finish our proof of Theorem 1, we enumerate these strips.

Proposition 2.30 (The number of strips of height m in W_{2n} is counted by the number $c_{2n,2m}$)**.**

$$s_{2n,m} = c_{2n,2m} = \binom{2n}{n-m} - \binom{2n}{n-m-1}$$

We show this by establishing a bijection between closed walks in W_{2n} with a specially marked strip of height m, and open walks of $2n$ steps starting at 0 and ending at height $2m$, the size of the latter being exactly $c_{2n,2m}$.

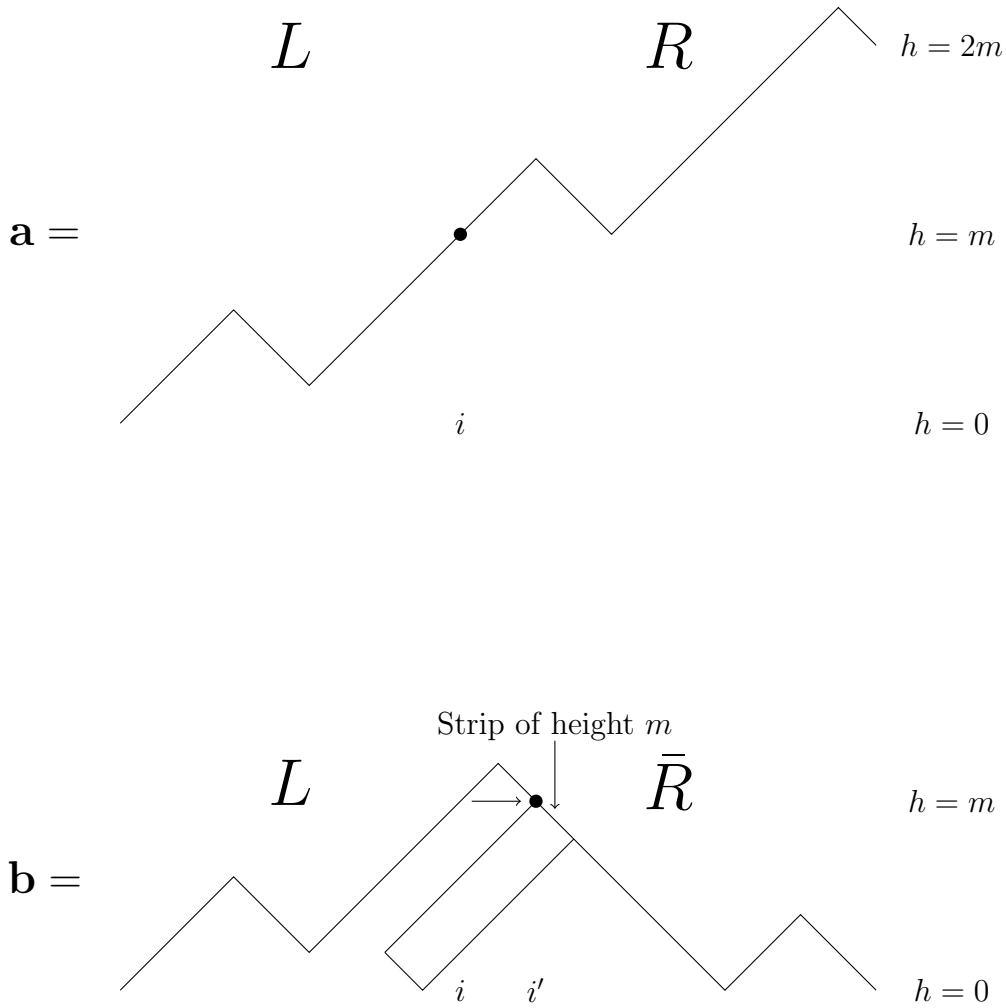


Figure 2.2: Example of the mapping from $a \in W_{2n}^{(2m)}$ to $b \in W_{2n}$ with a marked strip at height m. Here, we deal with the case where $h(i+1) \neq m-1$, and so we migrate $i \to i'$ where $i' = \min\{j > i | h(j) = m = h(j+1)+1\}$. We indicate this shift with an arrow, and also highlight the resulting strip of length m.

First, let us start with any walk $a \in W_{2n}^{(2m)}$. In this walk, the line $h = m$ must intersect a at an ascending slope at least once, where an ascending slope at a point j is defined by $h(j) = m$ and $h(j-1) = m-1$. Now, let i be the rightmost such point, namely, $i = \max\{j | h(j) = m \text{ and } h(j-1) = m-1\}$. Then, cutting a in half at the point i, we are left with two halves, L and R. The left half L may be viewed as an open walk $L \in W_i^{(m)}$, and the right half R may be viewed as an open walk $R \in W_{2n-i}^{(m)}$ by subtracting m from all of its heights and subtracting i from its step values, taking the interval $[i, 2n]$ to $[0, 2n-i]$. Note that by definition of i, the original R never falls below $h = m$.

Diagram showing this mapping

Then if we view R in the opposite direction, namely reflecting $R \to \bar{R}$ horizontally,

and then adjoining L and $\bar{R}$, i.e. connecting the end of L to the beginning of $\bar{R}$, we form a new closed walk $b = L\bar{R} \in W_{2n}$ with $h(i) = m$. Now, if in b we also have $h(i+1) = m-1$, then the box at point i is the end of a strip of height m, and we mark this box. If instead $h(i+1) = m+1$, then this box is not the end of a strip. However, we can simply take the next smallest point $i' > i$ such that $h(i') = m$ and $h(i'+1) = m-1$, and mark this topmost box at i'. This point always exists, as if $h(i+1) = m+1$, the walk must go back down to height 0 at position $2n$. This takes us from a walk $a \in W_{2n}^{(2m)}$ to a walk $b \in W_{2n}$ with a marked end of strip at height m.

Now, let us go the other direction. Starting with a walk $a \in W_{2n}$ with a marked end of strip at position i and height m, we must also have that $h(i+1) = m-1$. If we also have that $h(i-1) = m-1$, then i is a maximum of a, and it nicely separates a into halves L and R. L begins at $h(0) = 0$ and ends at $h(i) = m$, and so we have $L \in W_i^{(m)}$. R begins at $h(i) = m$ and ends at $h(2n) = 0$, and so reflecting $R \to \bar{R}$ horizontally, we have $\bar{R} \in W_{2n-i}^{(m)}$. Composing the two walks in the same fashion as above, we have $b = L\bar{R} \in W_{2n}^{(2m)}$, and since i was a maximum, we have $h(2n-1) = 2m-1$.

If instead $h(i-1) = m+1$, then i is not a maximum of a, but rather a descending slope. In this case, we find the largest point $i' < i$ such that $h(i') = m$ and $h(i'-1) = m-1$, i.e. the corresponding ascending slope to the left of i of height m. Proceeding with the same cutting, reflecting, and adjoining steps as previously described, we are then left with a walk $b = L\bar{R} \in W_{2n}^{(2m)}$. Here, since i' was an ascending slope, we have that $h(2n-1) = 2m+1$.

With this, we have defined a more specific mapping than promised:

(i) First, we have a mapping between closed walks $a \in W_{2n}$ with a marked end of strip of height m that is a maximum, and open walks $b \in W_{2n}^{(2m)}$ such that $h(2n-1) = 2m-1$, i.e. walks with a final ascending slope ending at height m.

(ii) Second, we have a mapping between closed walks $a \in W_{2n}$ with a marked end of strip of height m that is on a descending slope, and open walks $b \in W_{2n}^{(2m)}$ such that $h(2n-1) = 2m+1$, i.e. with a final descending slope ending at height m.

This establishes a bijection between closed walks $a \in W_{2n}$ with a marked end of strip of height m and open walks $b \in W_{2n}^{(2m)}$. Thus, we have shown that the number of strips of length m in all closed walks of length $2n$ is

$$s_{2n,m} = |W_{2n}^{(2m)}| = c_{2n,2m}, \tag{2.31}$$

and so we have proved Proposition 2.

In order to derive our original desired result stated in Theorem 1, we use our result from Proposition 2 (2.29) as well as our recently derived expression of the meander determinant in terms of the μs. Recall that $\mu_m = U_{m-1}/U_m$. Then, we have that

$$\det \mathcal{G}_{2n}(q) = \prod_{m=1}^{n} \left(\frac{U_m}{U_{m-1}}\right)^{s_{2n,m}} = \prod_{m=1}^{n} (U_m)^{s_{2n,m} - s_{2n,m+1}} \tag{2.32}$$

with the last equality justified by the cancellation encountered in subsequent terms. Further, note that when we reach the end of the product, we have $s_{2n,n+1} = \binom{2n}{-1} - \binom{2n}{-2} = 0$. Thus, we have exactly derived the equation stated in Theorem 1:

$$\det\left[\mathcal{G}_{2n}(q)\right] = \prod_{m=1}^{n} \left[U_m(q)\right]^{a_{2n,2m}}$$

$$a_{2n,2m} = c_{2n,2m} - c_{2n,2m+2} = s_{2n,m} - s_{2n,m+1}.$$

This completes our proof of Theorem 1.

Chapter 3

Further Exploration

To further explore the topic of meandric systems, we take an experimental mathematics approach, developing a toolbox with which to attack various questions regarding these objects, and hopefully to discover new patterns to be investigated more formally. Below, we will discuss our findings and provide a general overview and analysis of the structure of and the techniques used in the codebase. See the attached readme file and comments in the code for a more in depth explanation.

3.1 Meanders under Automorphism/Group Action

As discussed in La Croix 2003, meandric systems also exhibit certain symmetries under action from automorphisms. These symmetries can be used to cut out redundancies in the exhaustive construction of meandric systems of a certain size by considering only the representatives of the orbits under an action. We take advantage of this in our code to yield a constant time improvement in our calculation of the meander matrix.

The first and most obvious set of symmetries comes from rigid transformations of the plane. These consist of either reflecting along or across the line, or rotating 180° around a point. For all of these actions, any given element has an orbit of at most size 2, as a second reflection returns to the original configuration. Reflecting across the line is the most relevant of these for our purposes, as it is equivalent to swapping the top and bottom arch configurations, i.e. $\binom{a}{b} \longleftrightarrow \binom{b}{a}$. This yields a constant of 1/2 time improvement in our calculation of the meander matrix.

The other interesting symmetry comes from the symmetric group. This can be best visualized by considering the arch diagrams projected onto a sphere with a marked first point. Then, a rightward rotation is performed via a leftward rotation of the labels of the points. In terms of meandric systems in the plane, this can be visualized as a cutting of the rightmost connection, and a reattachment of that connection on the left side of the meandric system (see figure 3.1 below).

Any given meandric system has an orbit of size at most $2n$ under the symmetric action, but internal symmetries can be present to reduce the orbit size to a divisor of $2n$.

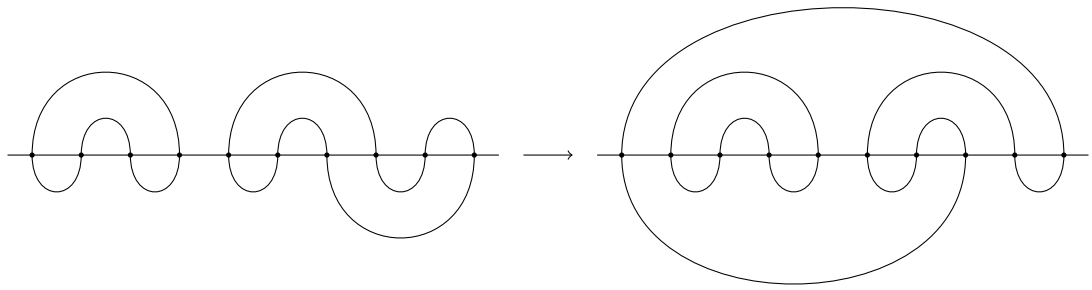


Figure 3.1: Rotation of a meandric system

From the across the line reflection and the symmetric action, we can also consider the dihedral action, which has orbit of at most $4n$. Note that for all of these actions, the number of connected components of the meandric system is fixed. Thus, any analysis of the connected components of one representative arch diagram extends by symmetry to all other arch diagrams in its orbit.

3.2 Experiments/Analysis

Analyzing the Gram matrices using both of these orderings yields interesting results. First, by assigning a brighter color to higher value cells, we yield graphics that contain patterns, some of which begin to look fractal-esque.

We generated meander matrices of up to size 8. Generating and analyzing these matrices seems to yield a good look into emergent patterns in meandric systems. However, larger matrices quickly become infeasible to generate, as the asymptotic growth of the Catalan numbers is given by

$$C_n \sim \frac{4^n}{n^{3/2}\sqrt{\pi}},$$

which is exponential in terms of n. Thus, a $C_n \times C_n$ matrix quickly becomes far too large to calculate or even store. One method that can be used to lessen this problem is to only calculate and store half of the matrix, as since it is symmetric, this retains all information. However, this only cuts the time to generate in half.

3.3 Meander Matrix Data

Given this large matrix of all of the meandric systems of a certain size, it is only natural to ask questions about certain statistics in the matrix. In general, our analysis seems to hint that many of the patterns we find in meandric systems are largely probabilistic and thus out of the scope of this paper, but this does not mean that combinatorial and heuristic analyses are useless. Instead, they can highlight areas of intrigue and avenues for further research.

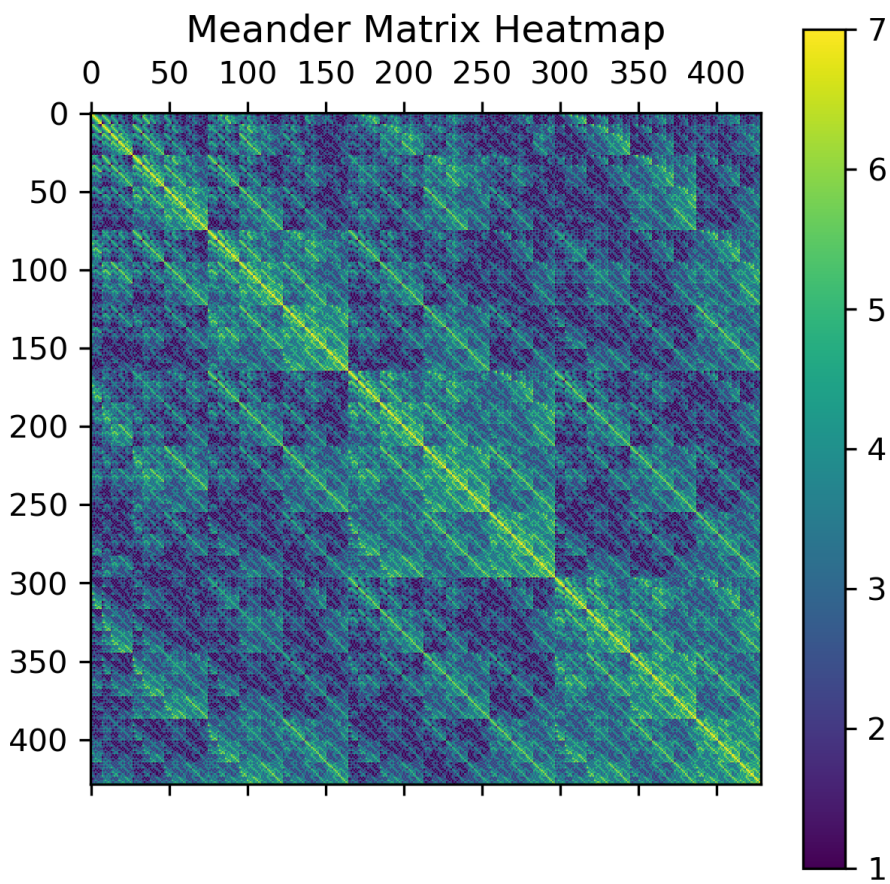


Figure 3.2: 8th Meander Matrix

N	Correlation	R^2	Correlation of Max Reps	R^2
3	-0.645	0.416	-0.999	0.998
4	-0.635	0.403	-0.982	0.964
5	-0.605	0.366	-0.958	0.918
6	-0.590	0.348	-0.926	0.857
7	-0.581	0.338	-0.909	0.826
8	-0.575	0.331	-0.860	0.740

Table 3.1: Correlation of height and component count of the n'th meander matrix, with corresponding R^2 values.

Any given row i of the meander matrix contains all of the meandric systems formed from arch diagram i. Thus, analyzing these rows can reveal general trends in the kinds of meandric systems formed. First, we found a statistically significant negative correlation between the height of an arch diagram and the number of components of its associated meandric systems. In other words, the greater the average distance covered by any given arch in an arch diagram, the less likely it is to form a higher number of connected components in associated meandric systems. This trend is much

more pronounced if one analyzes only representatives of arch diagram orbits, choosing the representative with maximum height. This is analogous to considering the arch diagram projected onto a sphere and choosing the starting point that results in the greatest height. [Heuristic analysis?]

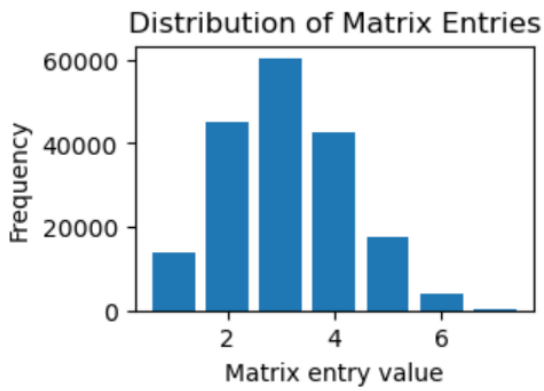


Figure 3.3: Distribution of the number of connected components in the 7th meander matrix

We also found that the distribution of component counts resembled a normal distribution (see figure 3.3).

Also, by looking at the minimum and maximum row sums for any given meander matrix, we come to the following two conjectures:

Theorem 3.1. For M_n, the arch diagrams with the minimum average number of connected components of their associated meanders are $u^n d^n$ and its rotations under the symmetric action. As such, there are n minimal arch diagrams.

Theorem 3.2. For M_n, the arch diagrams with the maximum average number of connected components of their associated meanders is $(ud)^n$ and its rotation $u(ud)^{n-1}d$. As such, there are always 2 maximal arch diagrams.

As discussed previously, the number of connected components of a meandric system is fixed under rotation. This is not precisely true of rotating just the upper arch diagram. However, given the average takes into account every associated meander, every possible pairing occurs and so the order of the counts of connected components is simply rotated within every orbit. This shows why the sets of extrema are composed of orbits. These conjectures also reflect the correlation between component count and height previously discussed, and why the correlation should be higher when considering maximal representatives.

3.3.1 Sequences of Meanders

In our search for patterns in meanders and the meander matrix, we encountered several sequences which, upon searching on OEIS, were met with a match. The following section details these experimental results and contains proofs for the claimed identities.

3.3.2 Maximally connected arch diagrams

As discussed above, we conjectured that the maximally connected arch diagrams are $(ud)^n$ and $u(ud)^{n-1}d$. Experimentally, we counted the total number of connects for this arch diagrams and gathered the beginnings of the sequence $1, 3, 10, 35, 126....$ Entering this into OEIS yielded the identity $\binom{2n-1}{n-1}$. Thus, we set our sights on finding a combinatorial proof for this.

Proposition 3.3. $F((ud)^n) = C_{n-1} + \sum_{j=1}^{n} C_{n-j}\binom{2j-3}{j-2} + C_{j-1}\binom{2n-2j-1}{n-j-1} = \binom{2n-1}{n-1}$

We will show this by induction on the first return of the upper Dyck word. First, let us assume that our identity holds for all $n \leq N$. Now, let us prove show it for $n = N + 1$.

First, let us partition according to first return. By connecting point 1 to point $2j$, we partition the counting problem into two parts, one under the arch and one to the right of the arch.

The section on the right is then of maximum size $n - 1$ if $j = 1$, and so is given by $\binom{2n-2j-1}{n-j-1}$ by our induction hypothesis. This quantity is added to the total sum of connected components once for every possible arch diagram it can be paired to the left of the first return. This yields the term $C_{j-1}\binom{2n-2j-1}{n-j-1}$.

Then, we note that the section under the arch is of maximum size $n - 1$ if $j = n$, and is equivalent to the Dyck word $u(ud)^{j-2}d$ by the transformation depicted in figure 3.4.

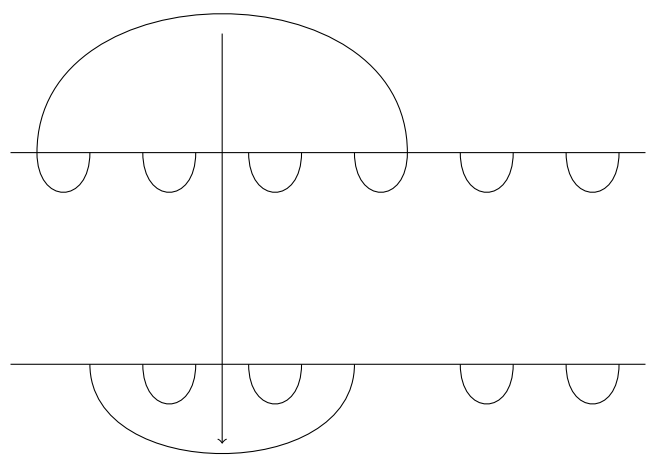


Figure 3.4: String pulling transformation

Recall that $F(u(ud)^{j-1}d) = F((ud)^j)$, and so the total number of connected components is given by $\binom{2j-3}{j-2}$. Similarly, this term is added to the total sum once for every possible arch diagram it can be paired with to the right of the first return. This yields the term $C_{n-j}\binom{2j-3}{j-2}$.

Now, note that when $j = 1$, $\binom{2j-3}{j-2}$ turns into $\binom{-1}{-1}$. This is nonsensical and classically defined to be 0. However, when $j = 1$, the first return immediately yields

one connected component, matching ud on top to ud on the bottom. Thus, we add a C_{n-1} term to account for every possible arch diagram of size $n-1$ which can occur to the right of the $j=1$ first return.

Now, all we have left to do is show a base case. For $n=1$, the only possible meandric system is the closed circle, yielding 1 total connected component. Thus, we are relieved to see that

$$C_0 + C_0\binom{-1}{-1} + C_0\binom{-1}{-1} = C_0 = 1.$$

Now, we show that $C_{n-1} + \sum_{j=1}^{n} C_{n-j}\binom{2j-3}{j-2} + C_{j-1}\binom{2n-2j-1}{n-j-1} = \binom{2n-1}{n-1}$. Defining the left term in the sum as A_j and the right term as B_j, we note that

$$A_{n+1-j} = C_{j-1}\binom{2n-2j-1}{n-j-1} = B_j.$$

Thus,

$$\sum_{j=1}^{n} A_j = \sum_{j=1}^{n} B_j$$

and so, noting that the $j=n$ case yields 0, we can rewrite the original formulation as

$$C_{n-1} + 2\sum_{j=1}^{n-1} C_{n-j-1}\binom{2j-1}{j-1} = \sum_{j=0}^{n-1} C_{n-j-1}\binom{2j}{j}$$

Now, note that $\binom{2n-1}{n-1}$ also counts NE lattice paths from $(0,0)$ to $(n-1,n)$. If we instead count these lattice paths by partitioning according to the first time the path goes above the diagonal, we get the sum

$$\sum_{j=0}^{n-1} C_j\binom{2n-2j-2}{n-j-1}.$$

C_j counts the number of ways to get to the diagonal at point j before a single up step is taken. Then, $\binom{2n-2j-2}{n-j-1}$ counts the number of NE lattice paths from $(j,j+1)$ to $(n-1,n)$. Now, by a natural rearranging of terms, we get that

$$\binom{2n-1}{n-1} = \sum_{j=0}^{n-1} C_j\binom{2n-2j-2}{n-j-1} = \sum_{j=0}^{n-1} C_{n-j-1}\binom{2j}{j},$$

and so we have our desired result, that the total number of components of meandric systems associated with $(ud)^n$ is given by $\binom{2n-1}{n-1}$. By rotational symmetry, this argument also applies to $u(ud)^{n-1}d$. □

This identity can also be proven from a different angle, highlighting another interesting pattern for these maximally connected arch diagrams. Let us define component sum polynomials as

$$G_n(q) = \sum_{a\in A_{2n}} q^{c(a,(ud)^n)}.$$

This polynomial encodes the number and order of all of the meandric systems associated with $(ud)^n$: if $G_n(q)$ contains the term rq^t, then there are r t-component meanders of size $2n$.

As we might expect, $G_n(1) = C_n$ since we are summing over all arch diagrams. More relevant to our previous work is that, generating the first few of these polynomials using our code base, we found that the polynomials generated exactly match what are called Narayana polynomials, denoted $N_n(q)$. This means that the number of meanders $\frac{a}{(ud)^n}$ with k connected components is given by the n, kth Narayana number, defined as

$$N(n,k) = \frac{1}{n}\binom{n}{k}\binom{n}{k-1}.$$

The first few Narayana polynomials are

$$\begin{aligned}
N_0(q) &= 1 \\
N_1(q) &= q \\
N_2(q) &= q^2 + q \\
N_3(q) &= q^3 + 3q^2 + q \\
N_4(q) &= q^4 + 6q^3 + 6q^2 + q \\
N_5(q) &= q^5 + 10q^4 + 20q^3 + 10q^2 + q
\end{aligned}$$

and Narayana polynomials are also known to satisfy the recurrence relation

$$N_n(q) = (1+q)N_{n-1}(q) + q\sum_{k=1}^{n-2} N_k(q)N_{n-k-1}(q). \tag{3.4}$$

Proposition 3.5. Maximally connected arch diagram component sum polynomials, defined above as $G_n(q)$ are precisely the Narayana polynomials $N_n(q)$.

Proof. Similar to the previous proof, we count via first return of the upper Dyck word at position $2j$. We know that $G_1(q) = q$, and let us define $G_0(q) = 1$. Then, for $j = 1$, we have a closed circle on the left, and on the right we have $n-1$ ud's left. Thus, this contributes the term $qG_{n-1}(q)$. Now, for $1 < j \leq n$, we use the same strand pulling argument as above (figure 3.4).

This gives us the term $G_{j-1}(q)G_{n-j}(q)$ for each j. Thus, we have the overall recurrence relation

$$G_n(q) = qG_{n-1}(q) + \sum_{j=2}^{n} G_{j-1}(q)G_{n-j}(q)$$

By pulling out the $j = n$ term and reindexing via $k = j+1$, we have the relation

$$(q+1)G_{n-1}(q) + \sum_{k=1}^{n-2} G_k(q)G_{n-k-1}(q).$$

This is precisely the recurrence relation and initial conditions satisfied by the Narayana polynomials. □

With this shown, it is only natural to assume that

$$G'_n(1) = \binom{2n-1}{n-1}$$

and this is exactly what happens in our code base. Now, we show it algabraically.

Proposition 3.6. $G'_n(1) = \binom{2n-1}{n-1}$

Proof.

$$\begin{aligned}
G'_n(1) &= \sum_{k=1}^{n} \frac{1}{n} k \binom{n}{k} \binom{n}{k-1} \\
&= \sum_{k=1}^{n} \frac{1}{n} k \frac{n!}{k!(n-k)!} \binom{n}{k-1} \\
&= \sum_{k=1}^{n} \binom{n-1}{k-1} \binom{n}{k-1} \\
&= \sum_{k=0}^{n-1} \binom{n-1}{n-1-k} \binom{n}{k} \\
&= \binom{2n-1}{n-1}
\end{aligned}$$

using Vandermonde's identity for the last step. □

Narayana numbers also count the number of Dyck words of length $2n$ with k peaks, as well as the number of noncrossing partitions of size n with k blocks. Notably, the corresponding arch diagrams to these constructs are not the same as the arch diagrams with n arches that form k connected components when paired with $(ud)^n$. Thus, the Narayana numbers seem to be counting something new here.

This also raises the question of what these polynomials might look like for arbitrary arch diagrams.

Proposition 3.7. Maximally connected arch diagram polynomials are greater than all other arch diagram polynomials for $x > 1$.

We graph the component sum polynomials of maximal orbit representatives for $n = 5$ in 3.7.

As hoped, the maximally connected arch diagram polynomials are greater than the other polynomials. This trend continues in our other testing.

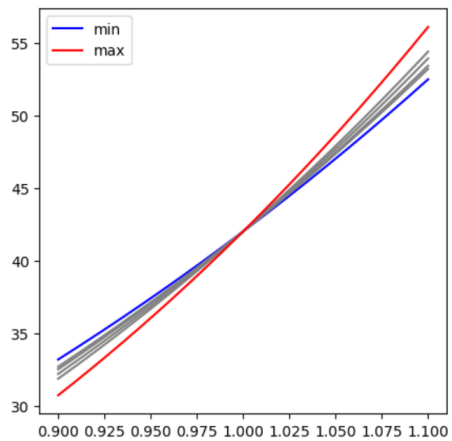


Figure 3.5: Polynomials centered around one, highlighting that the max has the highest slope

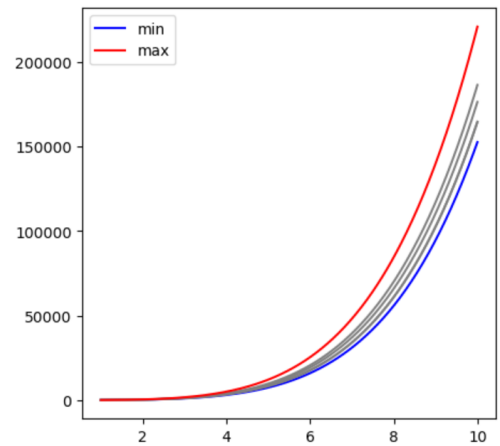


Figure 3.6: Polynomials in the range 0 to 10, highlighting that the conjectured maximum seems to be just that.

Figure 3.7: Component Sum Polynomials for $n = 5$

	Polynomial
min:	$10x + 16x^2 + 11x^3 + 4x^4 + x^5$
	$6x + 17x^2 + 13x^3 + 5x^4 + x^5$
	$8x + 15x^2 + 13x^3 + 5x^4 + x^5$
	$4x + 15x^2 + 15x^3 + 7x^4 + x^5$
	$6x + 14x^2 + 15x^3 + 6x^4 + x^5$
max:	$x + 10x^2 + 20x^3 + 10x^4 + x^5$

Table 3.2: Generated polynomial set with minimum and maximum elements.

Meander Macaronis

Previous papers have spent time enumerating certain subsets of meandric systems [see rainbow meander paper]. In a similar vein, we investigated the enumeration of what we have dubbed "meander macaronis". These consist entirely of structures with two arches on one side and two on the other, where the first side is two nested arches starting next to each other and ending next to eachother. These are closed with a single *ud* on each side. These upper and lower macaronis can be nested.

To simplify this, we note that each macaroni can be instead represented by a single arch. This turns the problem from counting meander macaronis on $2n$ points to counting interleavings of two Dyck words on $n/2$ points. By an interleaving, we mean the number of ways of fitting together the ends of two Dyck words represented as arches, where the two Dyck words are of orders a, b such that $a + b = n$. As such, we simply have to count the number of ways of choosing a and b and then count the number of Dyck words that can be built of those orders.

Proposition 3.8. In an order n system where $n = 2k$, the number of meander

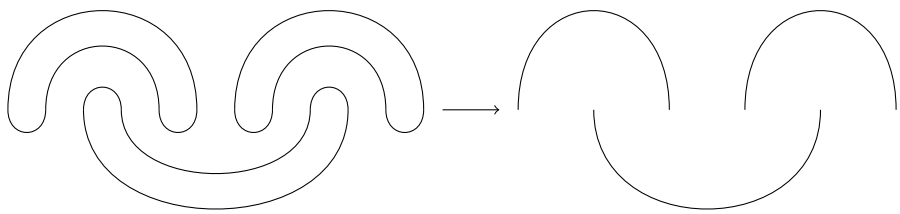


Figure 3.8: Representing a meander macaroni with arch diagram interleavings.

macaronis is given by

$$MM_n = \sum_{j=0}^{k} \binom{2k}{2j} C_j C_{k-j} = C_n C_{n+1}$$

The first equality of the proposition comes as follows: first, we note that in meandric macaronis, the ends of the arches correspond to pairs of points. Thus, we can assign the first two points the label 1, the third and fourth points the label 2, and so on. This leaves us with $2k$ points. From these, we can choose an even number of them for the top arch diagram, leaving an even number for the bottom arch diagram. This gives the $\binom{2k}{2j}$. Then, for the top points, we have C_j possible arch diagrams, and for the bottom points C_{k-j} possible arch diagrams.

The second equality we found by generating terms and entering the sequence into OEIS, which spat back out the above identity. This identity can be shown via rearranging terms, but this does not yield much intuition. The combinatorial proof for this is decently complex and is shown in [https://arxiv.org/abs/math/0601684]. □

3.3.3 Orbit related sequences

Below we include sequences involving the orbits of meanders derived from patterns in orbits of meanders.

n	Symmetric Action	Dihedral Action
1	1	1
2	2	2
3	7	5
4	30	18
5	188	99
6	1492	960

Table 3.3: Sequences of number of orbits over all meanders under the symmetric and dihedral actions

The above sequences did not garner any matches on OEIS, but we include them here regardless in case future endeavors produce the same sequence.

n	Symmetric Action	Dihedral Action
1	1	1
2	1	1
3	3	2
4	7	4
5	32	17
6	160	81

Table 3.4: Sequences of number of orbits over closed 1 component meanders under the symmetric and dihedral actions

3.4 Code

Throughout the code, we frequently rely on the Sagemath library, which contains implementations of many different mathematical objects of interest to us, such as Dyck words, permutations, and polynomials.

First, we need a way to represent arch diagrams. For this, we build arch diagrams from their associated Dyck words. We generate these Dyck words in their lexicographic ordering using the function

```
dyck_words(n, sorting = "lex").
```

This function employs simple recursion, wherein the recursive function calls itself twice, once with an open bracket and once with a close bracket. By keeping track of the number of opens and closes, the leaves at the end of the recursive branching will contain all Dyck words of length $2n$. At this point, they are added to the return list.

With these Dyck words we create a Meander class. From two Dyck words, a Meander can be created. To represent the arch diagrams, we use permutation representation with two cycles of the form $(a_1b_1)...(a_nb_n)$ where a_i is the index of the start of an arch and b_i is the end of that arch (i simply differentiates between arches for notation purposes here, and isn't present in the code). To create these, we use the existing tunnels() function for Dyck words, which returns a list of tuples of the form $(a_i, b_i + 1)$. We then store the arches as a list of tuples as described above, and also initialize top and bottom permutations using these lists. By multiplying the two permutations together, we get a permutation representing the whole meandric system. From this, we can extract the number of components (1.1).

With this, we are then able to implement the Gram matrix with respect to the bilinear form described previously (2.8) by entering at position (i, j) the value r, where r is the order of the meander $\frac{i}{j}$. Integer entries make the task of analyzing patterns in the matrix easier, but the entries can also be of the form q^r, which allows for the determinant discussed in Chapter 2 to be calculated. This uses Sage's polynomial ring implementation.

This process raises the question of a natural ordering for the Dyck words. The lexicographic ordering would appear to be the most obvious choice. By itself, it is easily generated recursively, as described above. Further, it turns out to be a refinement of the Tamari lattice, a well-studied partial ordering on Catalan objects.

As another heuristic here, we simply see more striking patterns in the meander matrix under the lexicographic ordering than under others. Overall, this choice does not matter immensely, but the lexicographic ordering seems to be the most natural.

The code does contain options for alternate orderings, including the Tamari lattice and the dominance ordering, also known as the Stanley lattice. The dominance ordering, as implemented by Sagemath, is a partial ordering defined by

$$a > b \text{ if } h(a) > h(b) \text{ at every point.} \tag{3.9}$$

We break ties first by comparing the maximum height of two Dyck words, and if these are equal, then the one with the leftmost first peak is less than the other one, as this means it is the first one to go down.

With all of these different orderings, the two methods

```
meander_matrix(n, sorting = "lex")

dyck_words(n, sorting = "lex")
```

take as input a sorting parameter, which is by default lexicographic, but can be changed to "dom" for dominance ordering or "tam" for default Tamari ordering.

With the meander matrix implemented, it is natural to ask if Di Francesco's formula for the determinant actually matches the calculated determinant, if only as a sanity check. Thus, we implement the formula using Sage's polynomial rings with a helper function calculating the constants defined in the paper, and quick tests show that it does indeed seem to work as hoped.

To implement the symmetric and dihedral actions, we create the function

```
rotate_dyck(x).
```

This uses the same tunnels() function as described previously. From this list of tuples, we make a new one where every up or down step is shifted one to the right, with the last step looping around to become the first. A Dyck word created from this new list of tuples is returned. For example,

```
x = DyckWord("()()()()")
y = rotate_dyck(x)
print(y)
>>> (()()())
```

Using this as the fundamental mechanism, we built several methods to analyze the orbits of meanders and arch diagrams.

```
orbit_indices(arches, index_dict, i, j = None)

meander_orbits(mM, arches)

arch_orbits(arches)
```

```
orbit_label_matrix(mM, arches)

get_reps(arches, orbits)
```

The orbit indices method takes as input the list of all arch diagrams of a certain size, a dictionary with Dyck words as keys and index of that dyck word in the arch list as entries, and either one or two indices. If it takes in one, it calculates the orbit of the arch diagram at that index under the symmetric rotation action. If it takes in two, then it calculates the orbit of the meander $\frac{i}{j}$ under the symmetric rotation action. In either case, it does this by rotating the involved Dyck words and adding the new index to an orbit set until the Dyck word(s) rotate back around to the original.

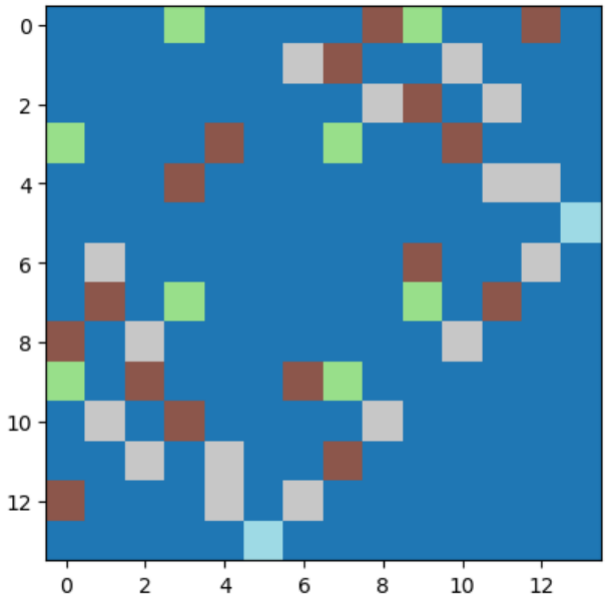
The meander orbits method takes as input a meander matrix of a certain size as well as the list of arch diagrams. It then builds an index dictionary as described above, and enumerates through all meanders in the matrix, calculating their orbits using the orbit indices method. By tracking which meanders have been visited, we do not double count. The arch orbits method does the same thing but just with single arch diagrams.

The orbit label matrix method finds all the meander orbits and creates a new matrix where each orbit is assigned a specific color in the matrix.

The get reps method takes as input the list of all arch diagrams and a list of orbits. It returns a list of representatives of the orbits. For meanders, it just returns any item as the representative. For arch diagrams, it returns the index of the arch diagram of maximal height in the orbit as the representative.

Here is an example of these methods in use.

```
# Lexicographically ordered meander matrix.
M, arches, heights = meander_matrix(4)
# Calculate orbits and label matrix accordingly
labels, num_orbits, orbits = orbit_label_matrix(M, arches)
# Print out plot of labels using Matplotlib library
plt.imshow(labels, cmap='tab20')
plt.colorbar()
plt.show()
```



The functions

```
component_sum_polynomial(n,x)

arch_orbit_polynomials(n)

arch_polynomial_graphs(n, x1, x2)
```

are used to analyze the component sum polynomials defined earlier in this chapter. The arch orbit polynomials function computes the component sum polynomials of a set of representative arch diagrams from the symmetric rotation orbits on arch diagrams. The arch polynomial graphs function calls the arch orbit polynomials function, and then graphs the polynomials with x values between $x1$ and $x2$.

The function

```
matrix_data(M, heights, should_print = true)
```

calculates and returns a variety of data about the meander matrix. First, it calculates the row sums of the matrix, and then with the list of heights it takes as input, calculates the correlation between the height of an arch diagram and the average number of connected components of its associated meanders. It also calculates the same correlation, but concerning only the maximum height representatives of arch diagram orbits under symmetric rotation. It returns these correlations along with a list of row sums, the indices of the minima and maxima of the row sums, and a list of indices of the arch diagrams ordered according to their heights.

The matrix data function also produces the following plots: row sums ordered according to row index along with a best fit line, a plot showing the distribution of row sums, a plot showing the distribution of matrix entries overall, and a matrix heatmap with brighter colors for higher values.

The code also contains the functions

```
draw_arch_diagram(dw)

draw_meander(top_dw, bot_dw)
```

which do as they say, creating matplotlib diagrams representing arch diagrams and meanders, respectively.

The code base can be found here: https://github.com/nikolaifriedberg/Thesis-stuff.

3.5 Conclusion

With the approach we use in our third chapter, we suspect there are more avenues to pursue and more results to be found. The methods used in the second chapter are also highly intriguing, and leave questions regarding the connection between Meanders, Chebyshev polynomials, and representation theory. We hope that this paper and the associated code inspires more research to be done on Meanders.

3.6 Bibliography

For my thesis, I am working with Kyle Ormsby on the meander problem. Discussed in a survey by Zvonkin 2024, the meander problem first arose in the early 20th century, though it often appeared in the disguise of some related question. They came under more scrutiny towards the end of the 20th century when Ed Witten constructed "certain invariants of 3-manifolds", which, in the coming years, were shown to only be invariants of the determinant of a certain matrix $\mathcal{M}(q)$ was equal to 0. This matrix turns out to be the "Meander Matrix", as discussed in depth in Ko and Smolinsky 1991Lickorish 1991. Deeper technical details may be hard to penetrate for this project, but some sources discussing these are Shum 1994Ayala and Francis 2024.

The main groundwork for my thesis is Di Francesco's proof that the meander matrix can be expressed as the Gram matrix of a change of basis in the Temperley Lieb AlgebraDi Francesco 1998. The equivalence of meanders and the Temperley Lieb Algebra opens up new avenues to be explored and gives new tools to work with. Further, the gram matrix is formed by a symmetric bilinear form, which opens up further avenues of study based on the equivalences with symmetric matrices. I will be using Kyle Ormsby's lecture notes on the Algabraic Theory of Quadratic FormsOrmsby n.d. and papers discussing techniques for factoring Chebyshev polynomialsGrubb 2014Trevisan and Wang 1998, which happen to be the othonormalizing coefficients for the Gram matrix basis. Hopefully by factoring these, we can better understanding the conditions under which the determinant of the matrix equals zero, and thus when Witten's proposed invariants apply.

Another angle of the project will be attempting to use nontrivial combinatorial approaches**la˙croix˙approaches˙nodate** to compute meander matrices for relatively large n. To compute these, the dyck paths equivalent to elements of the temperley lieb algebra will likely need to be sorted, methods for which are discussed in Daskalakis et al. n.d. The library SageMath will also be used to assist in this computational task *Sage - Dyck Words* 2025.

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Appendix A

The First Appendix

Appendix B

The Second Appendix, for Fun

thesis